



Context, data quality, and the root-cause diagnosis

Subscriber Churn Diagnosis · data & BI portfolio project · synthetic, seed-generated data  
Generated 2026-09-11

# Voxa+ — Subscriber Churn Diagnosis

A data & BI portfolio project. **Voxa+** is a fictional SVOD (subscription video-on-demand) streaming service operating in Brazil, Mexico, and the United States. Monthly subscriber churn nearly doubled starting in month 34 of a 36-month window and has stayed elevated. Leadership needs a data-driven root-cause diagnosis before the next board meeting.

This repository simulates the full analytics stack end to end: messy source exports, a medallion ETL pipeline, a Kimball star schema, a code-generated Power BI semantic model and report, and a dual-track (SQL + Python) root-cause investigation.

All data is synthetic and generated from a single random seed. Any resemblance to real companies or people is coincidental.

Visual case study (PDF) — 7-page carousel summary.

## Scope (locked in Phase 0)

|                      |                                                                                          |
|----------------------|------------------------------------------------------------------------------------------|
| Deliverable language | English                                                                                  |
| Data window          | 36 months; churn shift begins month 34                                                   |
| Markets / currencies | Brazil / BRL (home, month 1) · Mexico / MXN (month 10) · United States / USD (month 16)  |
| Reporting currency   | USD default, with a currency slicer (BRL / MXN / USD)                                    |
| Data scale           | ~120k cumulative subscribers, ~70k active at window end; viewing at subscriber-day grain |
| North Star           | Paid Active Subscribers · watch metric: Gross Monthly Churn %                            |
| Depth                | Balanced (data engineering + analysis)                                                   |
| PBI environment      | .pbip / PBIR, generated from code                                                        |

## Report pages (Phase 1)

- Executive Summary — how bad is the spike, is it still going
- Subscriber Retention & Cohorts — which cohorts / segments are driving it
- Content & Engagement — did viewing behavior drop before cancellation
- Pricing & Plans — did a price / plan change trigger it
- Acquisition Quality & Channels — did subscriber quality shift
- Customer Experience, Billing & App Quality — payment failures / app issues
- Market Context — global or concentrated in one market

## The diagnosis

The near-doubling of churn is **three overlapping factors**, none sufficient alone:

- an **app "v3" connected-TV playback regression** (the acute trigger — CTV-heavy churn 4.1% → 8.4% while the rest went 4.1% → 5.6%);
- the **Brazil price increase**, which did nothing to engaged subscribers and **+18 pp** to already-disengaged ones (the amplifier, Brazil-only);
- a **months-old Mexico cohort-quality drag** (newer Mexican cohorts retain ~8 pp worse at month 3 — the pre-existing drain).

Payments were tested and ruled out. Deliverables (built by `data_analysis/deliverables/build_deliverables.py`): board read — `board_summary.pdf` / `.html` · full write-up with all ten findings — `deep_dive.pdf` / `.html` (source: `deep_dive_00_index.md`).

## Repository layout

```
docs/          01-09 docs · 06_dax_measures & 03_data_dictionary auto-generated ·
               walkthrough.md (build log) ·
               build_docs_pdf.py → Voxa+_Documentation.pdf / Voxa+_Summary.pdf
etl/           config.py (one SEED) · gen/ world+content+economics · 01_generate_raw ·
               02_clean_stage · 03_build_curated · dq_checks · run_pipeline ·
               pbi_measures.py (113 DAX measures) · gen_data_dictionary / gen_dax_doc
data/          raw / staging / curated (Parquet + CSV mirror for compact tables) / quality
powerbi/       gen_semantic_model.py → VoxaChurnDiagnosis.SemanticModel (TMDL) ·
               gen_report_visuals.py + _drive_pbir.py + build_report.sh → 7-page report ·
               theme/theme.json · assets/ (brand PNGs)
assets/        brand.py (palette) · gen_logo.py (mark, wordmark, page backgrounds)
data_analysis/ run_analysis.py (10 questions × SQL + pandas, parity-checked) ·
               sql/ · results/ · parity/ · findings/F01-F10 · deliverables/
portfolio/     gallery/ (diagram + screenshot sources)
```

## Reproduce

---

```
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py                # raw → staging → curated → dq_checks (~28 min)
cd powerbi && bash build_report.sh         # semantic model + 7-page report (Desktop closed)
python data_analysis/run_analysis.py      # the dual-track diagnosis
python docs/build_docs_pdf.py             # the documentation PDFs
```

## Status

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Phases 0–12 complete — pipeline, semantic model, report, dual-track diagnosis and deliverables all build clean. Build log in [docs/walkthrough.md](#).



# Voxa+ — Business Context & KPI Framework

**Document status:** Phase 1 deliverable · authored at project kickoff, before any analysis. **Data generation date:** 2026-09-02 · **Analytics window:** 2023-09-01 → 2026-08-31 (36 full months). **Audience:** the Voxa+ leadership team and the board.

This document frames the problem and defines the metrics. It deliberately does **not** contain a conclusion about what caused the churn increase — that is the job of the investigation (Phase 10). Everything here was knowable *before* the diagnosis began.

## 1. The situation

Voxa+ is a subscription video-on-demand (SVOD) streaming service. For most of its history, monthly subscriber churn has run in a stable band. **Starting in month 34 of this window (June 2026), gross monthly churn nearly doubled and it has stayed elevated for the three months since.** Net subscriber additions have turned negative in the home market for the first time outside of a seasonal content gap.

Leadership has not been able to agree on a cause. Several things changed in the same broad period — pricing, the app, the content slate, the acquisition mix, the international footprint — and each function has a different theory. The board meets in five weeks and wants a single, evidence-backed diagnosis rather than seven competing narratives.

This project builds the data foundation and the analysis to deliver that diagnosis.

## 2. Company profile

| Name                | <b>Voxa+</b> (stylised "Voxa+"; legal entity <i>Voxa Mídia S.A.</i> )                                  |
|---------------------|--------------------------------------------------------------------------------------------------------|
| Headquarters        | São Paulo, Brazil                                                                                      |
| Business            | Direct-to-consumer SVOD streaming — films, series, and a growing slate of originals                    |
| Age                 | Operating in Brazil for ~6 years; the last 3 years (this window) cover the international expansion     |
| Employees           | ~640 (not modelled below the level of allocated G&A)                                                   |
| Fiscal calendar     | Calendar year; reporting is monthly                                                                    |
| Scale at window end | ~70k paid active subscribers, ~120k cumulative sign-ups across the window, ~US\$0.5M MRR (~US\$6M ARR) |
| Devices             | Mobile (iOS/Android), web, and connected-TV apps (Smart TV, streaming sticks, consoles)                |

### Positioning

Voxa+ is a **mid-priced general-entertainment** service — cheaper than the global majors, with a catalogue weighted toward Brazilian and Latin American content plus licensed international titles. Its edge is local originals and locally relevant pricing and payment methods (PIX and boleto in Brazil, OXXO and card in Mexico, card in the US).

## 3. The analytics window — why 36 months, and CY vs PY

| Anchor             | Value                                                               |
|--------------------|---------------------------------------------------------------------|
| Window length      | 36 complete months: <b>month 1 = Sep 2023</b> , month 36 = Aug 2026 |
| Current Year (CY)  | months 25–36 → <b>Sep 2025 – Aug 2026</b>                           |
| Prior Year (PY)    | months 13–24 → <b>Sep 2024 – Aug 2025</b>                           |
| Pre-shift baseline | months 1–33 (Sep 2023 – May 2026)                                   |
| Churn shift        | <b>begins month 34 (Jun 2026)</b> ; elevated in months 34, 35, 36   |

**Why 36 months.** It is the shortest window that contains the *entire* international expansion (both new markets launched inside it), gives a full CY-vs-PY comparison for the home market, and still leaves a long stable baseline before the shift so "normal" is well characterised. A shorter window would make the pre-shift band look like noise; a longer one would drag in a platform migration that predates the current data pipeline.

**Comparability caveat.** Because two markets launched mid-window, raw company-wide YoY mixes organic movement with expansion. Every time-series metric is therefore reported three ways: **Total**, **Comparable Base**, and **Expansion** (see §7).

## 4. Market & expansion timeline

| Market        | Currency   | Enters window        | Window-month | Notes                                                                                       |
|---------------|------------|----------------------|--------------|---------------------------------------------------------------------------------------------|
| Brazil        | BRL (R\$)  | present from month 1 | 1 (Sep 2023) | Home market. The only market present for the full window → the Comparable Base.             |
| Mexico        | MXN (MX\$) | Jun 2024             | 10           | Spanish-language localisation, OXXO cash payment, telco bundle partner from launch.         |
| United States | USD (US\$) | Dec 2024             | 16           | English UI, card-only, positioned against incumbents on price; smallest base, highest ARPU. |

Launch dates are modelled explicitly. Like-for-like analysis uses Brazil (full history) as the Comparable Base; Mexico gains its own 12-month YoY from month 22 (Jun 2025), the US from month 28 (Dec 2025).

## 5. Product catalogue, pricing, and the margin structure

### 5.1 Plan line-up (per market, monthly billing)

| Tier          | Description                                 | Brazil (R\$) | Mexico (MX\$) | US (US\$) |
|---------------|---------------------------------------------|--------------|---------------|-----------|
| Voxa Básico   | Ad-supported, 1080p, 1 stream, no downloads | 18.90        | 89            | 6.99      |
| Voxa Standard | Ad-free, 1080p, 2 streams, downloads        | 29.90        | 139           | 12.99     |
| Voxa Premium  | Ad-free, 4K HDR, 4 streams, downloads       | 44.90        | 219           | 17.99     |

- **Annual plans** are offered at all tiers for the price of 10 months (~17% discount), billed once and recognised monthly.
- A **7-day free trial** is available on Standard and Premium via direct sign-up (not via most partner/bundle channels).
- Prices shown are the line-up **as of window end**. The list price has not been static for the full 36 months — see §8 (events).

### 5.2 Revenue streams

| Stream           | ~Share of revenue | Mechanics                                                                                                                                           |
|------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Subscription     | ~86%              | Recurring plan fees, reporting-currency converted; annual plans amortised straight-line over 12 months.                                             |
| Advertising      | ~10%              | CPM-based, served against Básico-tier viewing. Revenue ≈ ad impressions × fill rate × eCPM. Seasonal (upfront-style H2 strength).                   |
| Partner / bundle | ~4%               | Revenue-share and wholesale arrangements with telcos (carrier billing in BR/MX) and one retail bundle. Lower net price, near-zero acquisition cost. |

### 5.3 Cost structure

| Cost                             | Nature        | Driver                                                                                                                                                                              |
|----------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Content — amortisation           | Largest cost  | Licensed titles amortised over licence term; originals over an accelerated useful-life curve. This is a CFO deep-dive thread (§9).                                                  |
| Content — cash spend             | Cash, lumpy   | Minimum guarantees and production milestones; diverges from amortisation month to month.                                                                                            |
| Streaming delivery (CDN)         | Variable      | ~per GB delivered; scales with viewing hours and bitrate (4K costs more per hour).                                                                                                  |
| Payment processing               | Variable      | % + fixed fee per attempt, <b>varies sharply by method</b> (card vs PIX vs boleto vs OXXO vs carrier billing). Also drives involuntary churn via failed renewals — CFO thread (§9). |
| Customer acquisition (marketing) | Semi-variable | Performance media + brand + partner bounties. Basis for CAC.                                                                                                                        |
| Customer support                 | Variable      | Cost per contact; contact rate per 1,000 subs.                                                                                                                                      |
| Cloud, platform & G&A            | Largely fixed | Allocated flat per subscriber for margin views.                                                                                                                                     |

### 5.4 The core tension

For a subscription business the revenue-vs-profit tension is **growth vs. retention economics**:

- Cheaper tiers (Básico) and partner/bundle channels **add subscribers fast and cheap** but carry lower ARPU, lower engagement, and historically weaker retention.
- Premium tier and direct sign-ups **retain and monetise better** but cost more to acquire and are a smaller share of net adds.
- Content spend lifts engagement and retention, but with a lag, and it is the line most visible to the board when margin is under pressure.

A churn increase is therefore never "just a retention problem" — it moves LTV, which moves the allowable CAC, which moves the growth plan. The investigation has to quantify the **profit and LTV impact**, not only the rate.

## 6. Business model — financial definitions and the North Star

### 6.1 Subscriber & churn definitions (the spine of this project)

| Term                                       | Definition                                                                                                                                                                            |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Paid active subscriber</b>              | An account with an in-force paid subscription on the measurement date (trials excluded until they convert).                                                                           |
| <b>New paid subscriber</b>                 | First paid billing in the period (trial conversion or direct paid start).                                                                                                             |
| <b>Reactivation</b>                        | A previously-churned account starting a new paid subscription.                                                                                                                        |
| <b>Gross subscriber churn (logo churn)</b> | $\text{Paid subscriptions that ended in the month} \div \text{paid subscriptions active at the start of the month}$ . <b>Primary watch metric</b> . Includes voluntary + involuntary. |
| <b>Voluntary churn</b>                     | Member-initiated cancellation (immediate or at period end).                                                                                                                           |
| <b>Involuntary churn</b>                   | Subscription ended because billing failed and the dunning sequence was exhausted.                                                                                                     |
| <b>Net subscriber adds</b>                 | $\text{New} + \text{reactivations} - \text{gross churn}$ .                                                                                                                            |
| <b>Net MRR retention (NRR)</b>             | $\text{MRR from a starting cohort 12 months later (incl. upgrades, downgrades, churn, win-back)} \div \text{starting MRR}$ . Secondary.                                               |
| <b>Cohort</b>                              | All accounts with the same first-paid month.                                                                                                                                          |
| <b>Mn retention</b>                        | Share of a cohort still paid-active n months after first paid billing.                                                                                                                |
| <b>Trial-to-paid conversion</b>            | $\text{Trials that convert to a first paid billing} \div \text{trials started}$ .                                                                                                     |

### 6.2 Revenue & margin definitions

| Term                        | Definition                                                                                                                                                |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MRR</b>                  | $\Sigma (\text{active paid subscriptions} \times \text{normalised monthly plan price})$ , reporting currency. Annual plans counted at 1/12.               |
| <b>ARPU</b>                 | $\text{Subscription revenue} \div \text{average paid active subscribers in the period}$ .                                                                 |
| <b>Total revenue</b>        | Subscription + advertising + partner/bundle.                                                                                                              |
| <b>Gross profit</b>         | Total revenue – content amortisation – CDN – payment processing – support.                                                                                |
| <b>Gross margin %</b>       | $\text{Gross profit} \div \text{total revenue}$ .                                                                                                         |
| <b>Contribution margin</b>  | $\text{Gross profit} - \text{marketing (acquisition} + \text{retention/lifecycle)}$ .                                                                     |
| <b>CM % (T12M)</b>          | $\text{Trailing-12-month contribution margin} \div \text{trailing-12-month revenue}$ .                                                                    |
| <b>Blended CAC</b>          | $\text{Total marketing spend} \div \text{new paid subscribers (incl. trial conversions, excl. reactivations)}$ . Scored <b>lower-is-better</b> .          |
| <b>LTV</b>                  | $\text{ARPU} \times \text{gross margin \%} \times \text{expected lifetime}$ , where $\text{expected lifetime} = 1 \div \text{monthly gross churn rate}$ . |
| <b>LTV : CAC</b>            | $\text{LTV} \div \text{blended CAC}$ .                                                                                                                    |
| <b>CAC payback (months)</b> | $\text{Blended CAC} \div (\text{ARPU} \times \text{gross margin \%})$ .                                                                                   |

### 6.3 North Star and watch metric

- **North Star: Paid Active Subscribers (end of period)**. Everything the company does is in service of a growing, healthy paid base.
- **Watch metric: Gross Monthly Churn %**. The North Star cannot grow if the base leaks faster than acquisition can refill it. This is the metric that broke in month 34.

Supporting the North Star: NRR (value retention), engagement health (leading indicator of churn), and LTV:CAC (whether growth is economic).

## 7. Total vs Comparable Base vs Expansion

| Cut                    | Definition                                                         | Use                                                                 |
|------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Total</b>           | All markets, all subscribers.                                      | Headline scale and the board number.                                |
| <b>Comparable Base</b> | <b>Brazil only</b> — the one market present for the entire window. | Like-for-like trend; strips expansion in/out of company-wide rates. |
| <b>Expansion</b>       | Mexico + United States.                                            | Isolates the contribution (and drag) of the new markets.            |

$\text{Expansion} = \text{Total} - \text{Comparable Base}$  for additive measures. Rates (churn %, ARPU, margin) are computed within each cut, never subtracted. Each market also carries its own `market_active_from` date so a per-market YoY only starts once 12 months of history exist.

An account-level flag `is_comparable_base` = (market = Brazil). A second flag `is_141_cohort` = (first-paid month ≤ month 24) supports 12-month cohort comparisons that are fully inside the pre-shift baseline.

## 8. Events in the window (neutral history)

These are things that happened. **None is asserted here to be a cause of the churn shift** — they are the candidate context the investigation will test, confirm, or rule out. Dates are approximate to the month.

| ~Month | Date         | Event                                                                                                                                          |
|--------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 6      | Feb 2024     | Cancellation flow redesigned (fewer steps, added a "pause subscription" option).                                                               |
| 10     | Jun 2024     | <b>Mexico launch</b> — OXXO payment, telco bundle partner live from day one.                                                                   |
| 13     | Sep 2024     | Support platform migration (new ticketing tool; contact tagging taxonomy changes).                                                             |
| 16     | Dec 2024     | <b>US launch</b> — card-only, price-led positioning.                                                                                           |
| 18     | Feb 2025     | Ad-tier eCPM deal renegotiated upward; ad load per hour increased slightly.                                                                    |
| 22     | Jun 2025     | Two long-running licensed international series expire from the Brazil catalogue.                                                               |
| 25     | Sep 2025     | Performance-marketing mix shifts toward lower-cost affiliate and paid-social inventory; a new telco bundle push begins in Brazil and Mexico.   |
| 28     | Dec 2025     | Annual-plan discount deepened for a holiday promo (echoes into renewals ~12 months later).                                                     |
| 30–33  | Feb–May 2026 | <b>Voxa+ app "v3" rollout</b> — re-platformed player and navigation, shipped device-family by device-family (mobile first, connected-TV last). |
| 31     | Mar 2026     | <b>Brazil list-price increase</b> on Standard and Premium; Básico held flat. Existing members migrated on their next renewal.                  |
| 32     | Apr 2026     | Plan-lineup tweak: Básico stream quality and ad load adjusted; downloads removed from Básico.                                                  |
| 34     | Jun 2026     | <b>Churn steps up.</b> <i>(This row is the symptom, not an event.)</i>                                                                         |

The investigation must weigh several of these against each other rather than seizing on the one nearest to month 34.

## 9. CFO-grade deep-dive threads

Two threads get modelled with extra rigour because they carry the profit and cash story:

### 9.1 Content economics

- **Amortisation vs cash spend** tracked separately; a monthly bridge between them.
- **Content cost per viewing hour** and **per retained subscriber**, by market.
- **Content supply cadence**: releases per month, days since the last "tentpole" title, and a share-of-catalogue-refreshed metric — the supply side of engagement.

### 9.2 Payments, dunning, and involuntary churn

- Payment **method mix** (card / PIX / boleto / OXXO / carrier billing) with method-level **authorisation rates**, **processing cost**, and **failure** → **recovery** funnels.
- **Dunning**: retry schedule, recovery rate by attempt, days-to-recovery, and the involuntary-churn tail when dunning is exhausted.
- Effect of the payment-method mix on **effective net price** and on **cash conversion** (carrier-billing settlement lag, boleto float).

Working-capital framing: subscription cash is collected close to service, so the classic cash-conversion-cycle story is thin; the cash levers here are **content cash timing** and **payment settlement lag**, reported as a simplified cash bridge rather than a full CCC.

## 10. Multi-currency handling

- Every monetary column is stored twice: `*_local` (transaction currency) and `*_usd` (converted at the month's average rate).
- A disconnected **Reporting Currency** table (BRL / MXN / USD) drives a slicer, with `FX Rate to Reporting Currency` (= 1 when reporting currency = USD) and a `Currency Symbol` measure ( R\$ / MX\$ / US\$ ).
- Leaf monetary measures are wrapped `× [FX Rate to Reporting Currency]`; **USD is the default** reporting currency for the portfolio.
- FX paths over the window are seeded with plausible drift: **BRL ≈ 4.9–5.5 / USD**, **MXN ≈ 17–20 / USD**. Rates move gradually; no single month swings enough to explain a churn step by itself, but FX is available as a control.

## 11. Seasonality and patterns embedded in the data

| Pattern                                              | Where it shows                                                                                               |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Dec–Jan</b> Brazilian summer + year-end tentpoles | Acquisition and engagement peak; annual-plan sign-ups spike.                                                 |
| <b>Carnaval</b> (Feb)                                | Short engagement dip in Brazil, ~1–2 weeks.                                                                  |
| <b>Jun–Jul</b> Festa Junina / winter                 | Engagement rises; a mid-year content push.                                                                   |
| <b>Content gaps</b> (weeks between tentpoles)        | Engagement troughs and a mild, <i>recurring, expected</i> churn uptick — the baseline seasonal churn rhythm. |
| <b>Annual-plan renewal echoes</b>                    | Sign-up peaks re-appear as renewal (and churn-decision) spikes ~12 months later.                             |
| <b>Payment cadence</b>                               | Boleto/PIX retries cluster at month start; carrier-billing subs bill and lapse on the partner's cycle.       |
| <b>Mexico</b>                                        | Semana Santa (Apr) and Día de Muertos (Nov/Dec) engagement bumps; December bundle promotions.                |
| <b>US</b>                                            | Post-holiday Q1 cancellations, a summer slump; short dip on Super Bowl weekend.                              |
| <b>FX drift</b>                                      | Gradual BRL/MXN depreciation vs USD nudges USD-reported ARPU down even when local ARPU is flat.              |

The month-34 shift sits **on top of** this seasonal structure and is larger than any seasonal move in the baseline.

12. Stakeholder personas → report pages

Each persona brings one standing question. Those questions are the seven Phase-1 pages.

| # | Persona                           | Standing question                                                                                                     | Page                                       |
|---|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 1 | CEO / Board                       | How bad is the spike, and is it still getting worse?                                                                  | Executive Summary                          |
| 2 | VP Retention & Lifecycle          | Which cohorts, tenures, plans, markets, and sign-up sources are actually churning?                                    | Subscriber Retention & Cohorts             |
| 3 | VP Content & Programming          | Did people stop watching <i>before</i> they cancelled? Is there a content-supply gap?                                 | Content & Engagement                       |
| 4 | VP Pricing & Monetization         | Did a price increase or plan change push people out — and how price-sensitive is each segment?                        | Pricing & Plans                            |
| 5 | VP Growth / Performance Marketing | Did the <i>quality</i> of who we acquire drop — worse channels, worse early retention?                                | Acquisition Quality & Channels             |
| 6 | VP Customer Experience / CTO      | Are payment failures or app problems forcing people out — involuntary churn, crashes, playback errors, support surge? | Customer Experience, Billing & App Quality |
| 7 | CFO / Head of Strategy            | Is this global or concentrated in one market — and what is the revenue and LTV hit?                                   | Market Context                             |

No single page is expected to contain the whole answer; the diagnosis comes from reading them together.

13. KPI framework

Benchmark ranges are **directional**, based on publicly disclosed SVOD metrics and analyst commentary, adapted to a mid-priced Latin-America-led service. They set the targets; they are not claims about competitors.

13.1 Growth & retention

| KPI                           | Definition                                 | Benchmark range | Target               |
|-------------------------------|--------------------------------------------|-----------------|----------------------|
| Paid Active Subscribers (EoP) | In-force paid accounts at period end       | —               | On plan curve (\$14) |
| Net Subscriber Adds / month   | New + reactivations – gross churn          | > 0 ex-seasonal | ≥ plan               |
| <b>Gross Monthly Churn %</b>  | Ended paid subs ÷ start-of-month paid subs | 3.0%–6.0%       | ≤ <b>4.0%</b>        |
| Voluntary Churn %             | Member-initiated share of churn            | 2.2%–4.5%       | ≤ 3.0%               |
| Involuntary Churn %           | Dunning-exhausted share of churn           | 0.7%–1.6%       | ≤ 1.0%               |
| Net MRR Retention (12M)       | Cohort MRR after 12M ÷ starting MRR        | 92%–102%        | ≥ 99%                |
| M1 Retention                  | Cohort active after first renewal          | 82%–90%         | ≥ 88%                |
| M6 Retention                  | Cohort active at month 6                   | 52%–70%         | ≥ 62%                |
| M12 Retention                 | Cohort active at month 12                  | 38%–55%         | ≥ 48%                |
| Trial-to-Paid Conversion %    | Converting trials ÷ trials started         | 45%–65%         | ≥ 55%                |
| Reactivation Rate %           | Win-back within 12M ÷ churned base         | 8%–16%          | ≥ 12%                |

13.2 Engagement (leading indicators of churn)

| KPI                                 | Definition                                                                            | Benchmark range | Target              |
|-------------------------------------|---------------------------------------------------------------------------------------|-----------------|---------------------|
| Monthly Active Rate %               | $\text{Paid subs who streamed} \geq 1 \text{ min in the month} \div \text{paid subs}$ | 74%–88%         | $\geq 85\%$         |
| Viewing Hours / Active Sub / month  | $\text{Streamed hours} \div \text{monthly active subs}$                               | 40–70 h         | 45–65 h             |
| % Subs Below Healthy Engagement     | Paid subs with < 5 streamed hours in the month                                        | 10%–20%         | $\leq 15\%$         |
| Titles Started / Active Sub / month | Distinct titles with a play start                                                     | 4–9             | $\geq 6$            |
| Days Since Last Tentpole Release    | Recency of a major release, by market                                                 | < 45 d          | $\leq 30 \text{ d}$ |
| Content Releases / month            | Titles added to catalogue                                                             | —               | $\geq \text{plan}$  |

### 13.3 Monetisation & unit economics

| KPI                           | Definition                                                     | Benchmark range                             | Target               |
|-------------------------------|----------------------------------------------------------------|---------------------------------------------|----------------------|
| ARPU (reporting currency)     | $\text{Subscription revenue} \div \text{avg paid subs}$        | BR ~US\$5.5–7 · MX ~US\$6–8 · US ~US\$11–14 | $\geq \text{plan}$   |
| MRR                           | Normalised recurring subscription value                        | —                                           | $\geq \text{plan}$   |
| Total Revenue YoY %           | Total revenue vs prior year (per cut)                          | —                                           | $\geq \text{plan}$   |
| Gross Margin %                | $\text{Gross profit} \div \text{revenue}$                      | 28%–45%                                     | $\geq 40\%$          |
| Contribution Margin % (T12M)  | $(\text{Gross profit} - \text{marketing}) \div \text{revenue}$ | 6%–18%                                      | $\geq 12\%$          |
| Blended CAC                   | $\text{Marketing} \div \text{new paid subs (lower is better)}$ | US\$14–30 blended                           | $\leq \text{US\$22}$ |
| CAC Payback (months)          | $\text{CAC} \div (\text{ARPU} \times \text{gross margin \%})$  | 12–20                                       | $\leq 16$            |
| LTV : CAC                     | $\text{LTV} \div \text{blended CAC}$                           | $\geq 3.0$                                  | $\geq 3.5$           |
| % Revenue from Ads Tier       | $\text{Advertising} \div \text{total revenue}$                 | 6%–14%                                      | context              |
| % New Subs via Partner/Bundle | $\text{Bundle sign-ups} \div \text{new subs}$                  | 10%–25%                                     | context              |

### 13.4 Acquisition quality

| KPI                                       | Definition                                                                | Benchmark range                         | Target               |
|-------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------|----------------------|
| New Subs by Channel                       | Direct / paid-search / paid-social / affiliate / partner-bundle / organic | —                                       | mix on plan          |
| Channel M1 Retention                      | M1 retention split by acquisition channel                                 | direct highest, affiliate/bundle lowest | each $\geq$ its plan |
| Channel M3 Retention                      | M3 retention split by acquisition channel                                 | —                                       | each $\geq$ its plan |
| Trial Start Rate by Channel               | $\text{Trials started} \div \text{sign-ups, by channel}$                  | —                                       | context              |
| Share of Adds from Low-Retention Channels | Affiliate + bundle share of new subs                                      | 20%–40%                                 | $\leq 35\%$          |
| Incentivised Sign-up Share                | Sign-ups with a promo/discount code                                       | 25%–45%                                 | $\leq 40\%$          |

### 13.5 Customer experience, billing & app quality

| KPI                           | Definition                                                       | Benchmark range                                                         | Target          |
|-------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------|
| Payment Failure Rate %        | $\text{Failed billing attempts} \div \text{attempts}$            | 4%–9%                                                                   | $\leq 6\%$      |
| Auth Rate by Method           | $\text{Approved} \div \text{attempted, per payment method}$      | card 88–95% · PIX 95–99% · boleto 70–85% · OXXO 65–80% · carrier 90–97% | per-method plan |
| Dunning Recovery Rate %       | $\text{Recovered} \div \text{entered dunning}$                   | 35%–55%                                                                 | $\geq 45\%$     |
| Days to Recovery              | Mean days from failure to successful charge                      | 2–7                                                                     | $\leq 5$        |
| App Crash-Free Session Rate % | $\text{Sessions with no fatal crash} \div \text{sessions}$       | 99.2%–99.8%                                                             | $\geq 99.5\%$   |
| Playback Error Rate %         | $\text{Streams with a fatal playback error} \div \text{streams}$ | 0.3%–1.2%                                                               | $\leq 0.7\%$    |
| Video Start Failure %         | $\text{Play attempts that never start} \div \text{attempts}$     | 0.5%–2.0%                                                               | $\leq 1.2\%$    |
| Support Contacts / 1,000 Subs | $\text{Tickets} \div (\text{subs} / 1,000) \text{ per month}$    | 40–90                                                                   | $\leq 70$       |
| CSAT                          | Mean post-contact satisfaction (1–5)                             | 3.9–4.5                                                                 | $\geq 4.2$      |

### 13.6 Market context

| KPI                                 | Definition                                           | Use                                          |
|-------------------------------------|------------------------------------------------------|----------------------------------------------|
| Churn % by Market                   | Gross monthly churn split BR / MX / US               | Is the shift global or concentrated?         |
| Net Adds by Market                  | Monthly net adds per market                          | Where the base is actually shrinking         |
| Revenue & MRR by Market             | Reporting-currency, with FX effect isolated          | Size of the financial hit                    |
| Engagement & App Quality by Market  | \$13.2 / \$13.5 metrics per market                   | Whether operational health differs by market |
| Cumulative Subscribers Lost vs Plan | (Plan base – actual base), by market, since month 34 | The board-level scoreboard of the gap        |

## 14. Targets and plan

- The plan is built on the **same basis as the actuals** (net of involuntary churn where the actual measure is; local-currency plan then FX-converted, not the reverse).
- **Plan subscriber base**:  $PY\ actual \times (1 + \text{planned growth})$  per market per month, with planned growth set *before* the shift (Brazil mid-single-digit YoY, Mexico and US ramping from their launch curves).
- **Plan churn**: flat at the \$13.1 targets (gross  $\leq 4.0\%$ , involuntary  $\leq 1.0\%$ ).
- **Plan ARPU / MRR**: plan base  $\times$  plan ARPU (local), then converted.
- **Plan marketing / CAC**: spend to hit plan net adds at  $CAC \leq US\$22$  blended.
- Plan is monthly for the full window; the CY plan (months 25–36) is the one shown against actuals on the report. "vs Target %" measures exist per KPI, with CAC and churn scored **lower-is-better**.

## 15. Out of scope (explicit)

- Title-level recommendation / personalisation algorithm analysis.
- Ad-tech yield and programmatic optimisation beyond blended eCPM and fill rate.
- Device- or OS-level quality-of-experience beyond crash rate, playback-error rate, and video-start-failure rate.
- Tax, transfer pricing, and statutory / IFRS reporting detail.
- Headcount, payroll, and opex below allocated G&A.
- A churn-**prediction** model. This project is a **diagnosis**, not a forecast.
- External data: competitor pricing, market share, third-party panel data. The "Market Context" page is internal markets (BR/MX/US), not the competitive market.
- Real personal data. All subscribers, payments, and viewing events are synthetic and seed-generated.

## 16. Resolved-assumptions log

| #   | Decision               | Resolution                                                                                                                                        |
|-----|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| A1  | Window length & anchor | 36 months, Sep 2023 – Aug 2026; data-gen date 2026-09-02.                                                                                         |
| A2  | Market launch months   | Brazil = pre-window; Mexico = month 10 (Jun 2024); US = month 16 (Dec 2024).                                                                      |
| A3  | Churn-shift timing     | Steps up at month 34 (Jun 2026); elevated for months 34–36.                                                                                       |
| A4  | Primary churn metric   | Gross monthly logo churn; voluntary/involuntary split reported alongside.                                                                         |
| A5  | North Star             | Paid Active Subscribers (EoP); watch metric = gross monthly churn %.                                                                              |
| A6  | Plan lineup            | Three tiers (Básico ad-supported / Standard / Premium), monthly + annual, per-market local pricing.                                               |
| A7  | Trial                  | 7-day free trial on Standard/Premium via direct sign-up only.                                                                                     |
| A8  | Comparable Base        | Brazil only (sole full-window market); Expansion = Mexico + US.                                                                                   |
| A9  | Reporting currency     | USD default; slicer for BRL/MXN/USD; <code>*_local</code> + <code>*_usd</code> on every money column.                                             |
| A10 | FX method              | Monthly average rates, seeded gradual drift; no single-month FX shock.                                                                            |
| A11 | Content accounting     | Amortisation and cash spend modelled separately, with a monthly bridge.                                                                           |
| A12 | Payment scope          | Method mix, auth rates, dunning funnel, involuntary-churn tail modelled; full CCC not.                                                            |
| A13 | Fiscal calendar        | Calendar year; monthly reporting.                                                                                                                 |
| A14 | Benchmark ranges       | Directional, from public SVOD disclosures/analyst ranges; used only to set targets.                                                               |
| A15 | Root cause             | <b>Deliberately not decided in this document.</b> Determined during ETL (Phase 3), surfaced only through the data and the Phase 10 investigation. |



# Voxa+ — Data Quality

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**Document status:** Phase 4 deliverable. **Generated companion:** `data/quality/dq_report.md` (rebuilt every pipeline run by `etl/dq_checks.py`), fed by `data/quality/dq_fragments/*.json`.

Every issue in the Phase-2 catalogue (Q01–Q15) is **injected** by `01_generate_raw.py`, **handled** in staging or curated, **counted** into a DQ fragment, and — where it is an invariant — **asserted** by `dq_checks.py`. Counts below are from the shipped full-scale run ( `SEED=20260902`, `POP_SCALE=1.0` ).

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## 1. Issue → handling → evidence

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| #   | Issue                                                                        | Injected (count)                                                            | Handled in                                      | Handling                                                                                                                           | Asserted                                                                   |
|-----|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Q01 | Multiple date formats + Excel serials                                        | 47 serial cells + all date columns                                          | 02 parse_date_series                            | per-style to_datetime ; serial = EXCEL_EPOCH + days ; 125,638 serials converted, 140,024 text dates parsed                         | dates_unparseable = 0                                                      |
| Q02 | Money as locale strings, parentheses negatives                               | 1,173,363 cells                                                             | 02 parse_money_series                           | decimal separator decided per row; (x) → -x (10,190); 1,161,624 parsed                                                             | no money column left as object dtype                                       |
| Q03 | Encoding damage (Latin-1 mojibake, BOM)                                      | 114,784 names + 3,500 titles + 1 BOM                                        | 02 ftfy.fix_text , utf-8-sig , Latin-1 read     | 117,061 strings repaired, BOM stripped                                                                                             | no Ã/Â runs in dim_content.title                                           |
| Q04 | Duplicate rows (events, invoices, daily full-dump)                           | 4,771 + 303,064 + 303,064 events; 32,248 invoice; 2,442,218 snapshot copies | 02 drop_duplicates ; per-file snapshot collapse | 623,966 dup events dropped; 32,310 invoice lines; 2,442,218 snapshot rows collapsed                                                | fact_subscription_event unique on event_id ; invoices unique on invoice_id |
| Q05 | Master rows missing (orphan subscriber / content)                            | 465 GDPR-deleted + 101,533 orphan content                                   | 03 sentinel keys -1 / 0 + count                 | 1,025 event rows, 8,003 billing rows, 101,533 viewing rows → sentinel                                                              | every fact FK resolves (real or sentinel); no null surrogate keys          |
| Q06 | Mislabelled columns ( clicks =impressions; spend_usd =MXN; amount gross/net) | 336 + 648 + 478,434                                                         | 02 column correction; 03 re-derive amount       | partner clicks →null, MX spend restored to MXN + USD recomputed (984 corrected); billing amount taken from the plan-price schedule | marketing clicks ≤ impressions                                             |
| Q07 | Snapshot-vs-ledger drift                                                     | 14,652 drift rows                                                           | 03 snapshot folded from the ledger              | Ketch snapshot used only to measure drift; curated status = ledger status by construction                                          | base roll-forward identity holds (worst 0.00 % monthly)                    |
| Q08 | Sign errors (negative refunds, negative watch, negative recovery)            | 11,546 + 33,569                                                             | 02 sign normalisation                           | refunds stored with consistent sign; 33,569 negative watch rows → abs + flagged                                                    | fact_viewing_daily.streamed_minutes ≥ 0                                    |
| Q09 | Unit inconsistency (watch s→ms, CDN MB/GB, CSAT /5 /10)                      | 12,954,757 + 6,736,491 + 79,380                                             | 02 unit normalisation                           | ms era + magnitude → seconds; edge-b bytes ÷ 1024; CSAT rescaled to 0-1                                                            | watch minutes/sub-day ≤ 1440; csat_score ∈ [0,1]                           |
| Q10 | Categorical noise (channel/plan/market/device free text)                     | ~1.77 M values touched                                                      | 02 norm_series + etl/vocab/*.csv                | 1,766,151 values normalised; 0 fell through to unknown this run                                                                    | all dim_* category columns ∈ controlled vocab                              |
| Q11 | Schema break (Helpline v1 → Helpdesk v2)                                     | 5,344 v1 + 74,036 v2; 164 cutover dupes                                     | 02 schema unification + crosswalk               | one stg_support_tickets schema + one taxonomy; 165 cutover dupes dropped                                                           | fact_support_ticket single schema; taxonomy ∈ dim_support_reason           |
| Q12 | Late / partial / restating files                                             | 4 FX months missing; last month partial                                     | 02 FX forward-fill; dim_date.is_partial_period  | 4 FX months filled; window-month 36 flagged partial                                                                                | last-month completeness flagged, not silently averaged                     |
| Q13 | Null keys (guest billing rows, retired plan)                                 | 3,424 null account_id                                                       | 02 flag; 03 sentinel                            | 3,424 flagged; billing rows with unresolved subscriber = 0.70 %                                                                    | soft check ≤ 1.2 %                                                         |
| Q14 | Outliers (test campaign, 30h+/day, annual as one line)                       | test-campaign flag; annual invoices                                         | 03 quarantine / explode / caps                  | annual plans explode to 12 monthly recognitions; test campaign excluded from fact_marketing_spend                                  | marketing KPIs exclude the quarantined campaign                            |
| Q15 | Timezone / month-boundary bleed                                              | UTC events vs local billing                                                 | 02 local-day derivation per market              | local_date from tz-converted timestamp; month_idx from the local day                                                               | monthly viewing totals reconcile to daily within tolerance                 |

## 2. Hard checks (fail the build)

From the latest run — **6/6 PASS**:

| Check                                                                                                                                    | Result                              |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| No null subscriber_key in fact_subscription_month                                                                                        | PASS                                |
| fact_subscription_event subscriber_key resolves or sentinel                                                                              | PASS                                |
| Subscriber base roll-forward identity $\text{start} + \text{new} + \text{reactivation} - \text{churn} = \text{end}$ ( $\leq 2\%$ /month) | PASS — worst 0.00 %                 |
| Cohort retention $\leq 100\%$ and non-increasing                                                                                         | PASS — M1 89 % / M6 74 % / M12 64 % |
| fact_viewing_daily.device_key $\subseteq$ dim_device                                                                                     | PASS                                |
| dim_date spans beyond the fact window                                                                                                    | PASS                                |

## 3. Soft checks (plausibility & calibration — warn only)

From the latest run — **0 warnings**:

| Check                                                    | Value  | Band         |
|----------------------------------------------------------|--------|--------------|
| Billing rows with unresolved subscriber                  | 0.70 % | $\leq 1.2\%$ |
| Pre-shift blended gross monthly churn (idx 24–32)        | 4.11 % | 3.6–4.6 %    |
| Peak blended gross churn (idx 33–35)                     | 6.82 % | 6.4–8.6 %    |
| Peak / pre ratio                                         | 1.66×  | 1.6–2.1×     |
| Avg streamed hours / sub-month (pre-shift)               | 47.6 h | 25–70 h      |
| Share below healthy engagement (pre-shift)               | 11.1 % | 8–28 %       |
| GL subscription revenue vs curated MRR (mean rel. error) | 4.0 %  | $\leq 15\%$  |

The full churn-spike shape and cohort-retention curve are tabulated in `data/quality/dq_report.md`.

## 4. Accepted limitations

These are known and left in on purpose; the report and the Phase-10 analysis annotate them.

- Acquisition channel is a sampled attribute.** The AdBridge attribution feed carries no account key, so each subscriber's `acquisition_channel` is drawn from their market's monthly channel mix. Channel-level retention and quality trends are directionally sound but not row-exact; conclusions about acquisition quality are stated as cohort-level, not individual-level.
- Billing amount is taken from the plan-price schedule**, not the gateway `amount` column (Q06 mislabel). Revenue and payment-cost figures are therefore schedule-accurate; the gateway's reported amount is kept as `amount_reported` for audit only.
- The last window month is partial.** `dim_date.is_partial_period` flags window-month 36; partner settlement and some carrier-billing rows lag by a month. Trend visuals should exclude or annotate it.
- Snapshot-vs-ledger drift is measured, not corrected.** `fact_subscription_month` is folded from the movement ledger; the 14,652 drifting Ketch snapshot rows are reported as a DQ metric, not reconciled.
- fact\_finance\_month is a modelled roll-up**, not a general ledger. It ties to operational MRR within  $\sim 4\%$  and is adequate for margin trend and the CFO page, not for statutory reporting.
- M12 cohort retention ( $\sim 64\%$ ) sits a little above the docs/01 benchmark band (38–55 %).** The simulated base carries a  $\sim 15\text{--}19\%$  annual-plan mix (locked in) and counts win-backs as re-activations, both of which lift 12-month logo retention. The Phase-10 analysis uses first-spell-only retention where a stricter figure is needed.
- Gross margin is thin and ramps.** Content amortisation is  $\sim 90\%+$  of revenue early in the window on a small base and settles near  $\sim 55\%$  by window-end (26 % gross margin at window-month 34) — a deliberate scaling-company trajectory, slightly below the docs/01 steady-state band.
- No external / competitor data.** The "Market Context" page compares Voxa+'s own three markets, not the competitive market.



# Voxa+ — Subscriber Churn: Root-Cause Diagnosis

Analyst deep-dive · prepared for the leadership team · all figures dual-computed in DuckDB SQL and pandas with a parity check on every result ( `../parity/parity_report.md` — 14/14 pass).

## 0. The question and the answer

**Question.** Gross monthly subscriber churn nearly doubled starting at window-month 34 (Jun 2026) and has stayed elevated. What caused it?

**Answer.** Three factors that overlapped in time. None explains the spike on its own.

|   | Factor                                                                                                     | Role                               | Markets                |
|---|------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------|
| 1 | An app "v3" <b>connected-TV playback regression</b> (window-month 32)                                      | acute trigger, biggest contributor | all; weakest in the US |
| 2 | The <b>Brazil Standard/Premium price increase</b> (window-month 32), landing on an already-disengaged base | amplifier                          | Brazil only            |
| 3 | A <b>months-old Mexico cohort-quality drag</b> (newer Mexican subscribers retain ~8 pp worse)              | pre-existing drain                 | Mexico only            |

Payments and involuntary churn were tested and are **not** involved ( §6 ).

## 1. The spike — size, persistence, shape (*findings F01, F02*)

Gross monthly churn stepped from a **4.1 %** pre-shift baseline (window-months 25–33) to **6.8 %** across the three elevated months — a **1.66×** increase, and **90 % of it is voluntary**. It has not receded: the three months read 6.85 %, 6.58 %, 7.02 %.

Splitting by market shows the company-wide number is a composite of three different shapes:

| Market        | Pre-shift | Peak  | Behaviour                                                                                          |
|---------------|-----------|-------|----------------------------------------------------------------------------------------------------|
| Brazil        | 3.4 %     | 7.0 % | flat, then a clean <b>+3.0 pp step</b> at window-month 34 — acute                                  |
| Mexico        | 5.4 %     | 8.1 % | <b>drifting up from ~window-month 29</b> (+0.6 pp), then a further step — a problem that got worse |
| United States | 4.1 %     | 4.4 % | essentially unchanged — the control                                                                |

A US that barely moves already rules out any explanation that would hit every market equally.

## 2. Who is leaving, and why now (*findings F03, F04*)

**The lift is broad, not narrow.** Peak-vs-baseline churn is up by a strikingly even ~+2.7 pp across almost every tenure band, plan tier and acquisition channel. Two exceptions carry information:

- **Annual plans lifted only +1.9 pp** (vs +2.7 pp monthly) — locked-in subscribers were partly shielded.
- **Long-tenure subscribers (25+ months) were hit as hard as brand-new ones** (+3.2 pp) — a cause confined to weak new cohorts could not do that.

A single narrow cause does not produce an even cross-segment lift. This is the first quantitative sign of **multiple overlapping factors**.

**The extra churn is engagement-led.** For subscribers who cancelled during the elevated months, streamed hours were already sliding beforehand — ~37 h two months out, ~36 h the month before — whereas pre-shift churners left from a flat ~43 h. Value erodes first, the cancellation follows. That points at what suppressed viewing just before the spike (§3), and it explains why a price change bites harder now (§4): people were already getting less.

## 3. Factor 1 — the connected-TV app regression (*finding F05*)

The Voxa+ "v3" app shipped device-family by device-family, reaching connected-TV last. Connected-TV streaming quality is flat through window-month 31, then:

| window-month                 | 31    | 32           | 33–36  |
|------------------------------|-------|--------------|--------|
| CTV video-start-failure rate | 1.1 % | <b>3.0 %</b> | ~3.4 % |
| CTV playback-error rate      | 0.6 % | <b>1.4 %</b> | ~1.6 % |

Mobile and web are unchanged throughout. The connected-TV rates **triple at window-month 32 — one month before the churn step**.

The churn consequence is specific to the affected subscribers:

|                                                                | Pre-shift | Peak         |
|----------------------------------------------------------------|-----------|--------------|
| Not CTV-heavy                                                  | 4.1 %     | 5.6 %        |
| <b>CTV-heavy</b> (primary device = smart-TV / stick / console) | 4.1 %     | <b>8.4 %</b> |

Before the shift the two groups churned **identically**. After the CTV rollout, CTV-heavy churn **doubled**. The one-month lag between the quality drop and the churn step is a value → decision delay of roughly one billing cycle. This is the **acute trigger and the largest single contributor**, and — because it touches every market — it is why the company-wide number looks global.

#### 4. Factor 2 — the Brazil price increase, and why it bit (*finding F06*)

Brazil raised Standard and Premium list prices at window-month 32, migrating existing members on their next renewal. Split Brazil Std/Prem subscribers by whether they were engaged or disengaged (below the healthy-engagement line) in the month:

| Brazil Std/Prem   | Churn before hike | Churn after hike | Lift            |
|-------------------|-------------------|------------------|-----------------|
| <b>Engaged</b>    | ~0 %              | ~0 %             | ≈ 0 pp          |
| <b>Disengaged</b> | 23.9 %            | <b>42.2 %</b>    | <b>+18.3 pp</b> |

The price increase had **no measurable effect on engaged subscribers** and a very large effect on **already-disengaged** ones. Because Factor 1 pushed a wave of Brazil subscribers into the disengaged bucket in window-months 32–33, the migration then landed on a larger disengaged pool than it otherwise would have. Price alone did not move the base; **price × prior disengagement produced the Brazil step**. Brazil-only.

#### 5. Factor 3 — Mexico's pre-existing cohort-quality drag (*finding F07*)

Month-3 retention by market and cohort era:

| Market        | Pre-shift cohorts | Post-shift cohorts (acquired m24–30) |
|---------------|-------------------|--------------------------------------|
| Brazil        | ~82.8 %           | ~80 % (–3 pp)                        |
| <b>Mexico</b> | ~81.8 %           | <b>~73 % (–8–9 pp)</b>               |
| United States | ~82.8 %           | ~81 % (–2 pp)                        |

Every market's newer cohorts retain a little worse; **Mexico's are ~8–9 pp worse at month 3** — three times the drop elsewhere. It is **not channel-specific** (within post-shift Mexico, low-quality-channel cohorts retain about the same as the rest), so this is not "the channel mix kept degrading" — it is **Mexican conditions for newly-acquired subscribers deteriorating across the board** from ~window-month 24. This drag was already lifting Mexican churn ~0.6 pp before the visible spike and is still underneath it.

#### 6. What it is *not* — payments (*finding F08*)

| Market        | First-attempt payment-failure rate (pre → peak) | Dunning recovery (pre → peak) | Involuntary churn (pre → peak) |
|---------------|-------------------------------------------------|-------------------------------|--------------------------------|
| Brazil        | 7.6 % → 7.7 %                                   | 51 % → 52 %                   | 0.43 % → 0.41 %                |
| Mexico        | 18.8 % → 18.2 %                                 | 37 % → 37 %                   | 1.25 % → 1.30 %                |
| United States | 8.0 % → 7.8 %                                   | 58 % → 57 %                   | 0.31 % → 0.32 %                |

Nothing moved. Combined with the 90 % voluntary share (\$1), payments are **ruled out** as a driver of the spike. Mexico's payment economics are structurally poor but that is a standing cost, not a new one.

#### 7. Impact, and the decomposition (*findings F09, F10*)

**Impact.** The churn step cuts the implied average subscriber lifetime from ~24 months to ~15 — a ~40 % reduction in lifetime value. In the three elevated months, ~5,400 subscribers left over and above the pre-shift rate; at ~\$7.5 ARPU that is on the order of **\$0.5–0.6 M of lifetime recurring revenue already forgone**, compounding monthly. A 40 % lower LTV also cuts the allowable CAC by ~40 %, so the growth plan no longer pencils.

**Decomposition.** Tag every peak-month voluntary churn with the three factor flags:

| Market        | Peak voluntary churn | F1 (CTV) | F2 (price×diseng) | F3 (low-q young) | none        |
|---------------|----------------------|----------|-------------------|------------------|-------------|
| Brazil        | 6,629                | 54 %     | <b>63 %</b>       | 12 %             | <b>15 %</b> |
| Mexico        | 4,017                | 53 %     | 0 %               | <b>23 %</b>      | 36 %        |
| United States | 1,597                | 52 %     | 0 %               | 20 %             | 39 %        |

In Brazil, **85 % of the elevated churn carries Factor 1 or Factor 2**. In Mexico, ~64 % carries Factor 1 or Factor 3. The US "39 % none" is just its unchanged ~4 % baseline — the control. The flags overlap, so these are *coverage*, not an independent split; the causal weight sits in the **rate** comparisons (\$3–\$5), which are clean.

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## 8. Recommendations (priority order)

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1. **Fix the connected-TV player (Sev-1)**. Roll back / hotfix v3 on smart-TV and streaming-stick; verify video-start-failure returns to ~1 %. Save-campaign CTV-heavy subscribers whose viewing fell after window-month 32.
  2. **Pause the Brazil price migration for low-engagement Standard/Premium subscribers**. Offer a retention path (hold price / downgrade / pause); re-test elasticity after the CTV fix.
  3. **Audit Mexico acquisition and onboarding since mid-2025**. Cap spend on the weakest sources until month-3 retention recovers toward ~82 %.
  4. **Instrument for next time**. Connected-TV QoE on the executive dashboard, a leading "engagement health" trend, and price-change elasticity by engagement band.
- 

## How this was produced

`python data_analysis/run_analysis.py` runs all ten questions twice — DuckDB SQL over the curated Parquet and pandas over the same files — and asserts the two agree within tolerance before any number is used. SQL is saved under `data_analysis/sql/`, agreed results under `data_analysis/results/`, and the parity log at `data_analysis/parity/parity_report.md`. Per-problem method, evidence and limitations are in `data_analysis/findings/F01 – F10`.